

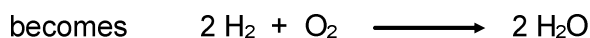
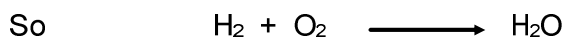
Balancing Equations

Law of Conservation of Mass

During a chemical reaction, the total mass of the reacting substances (*reactants*), is always equal to the total mass of the resulting substances (*products*).

In other words:

- mass is not lost or destroyed
- the total mass of reactants must equal the total mass of products
- the number of atoms of reactants must equal the number of atoms of products



There are 4 hydrogen atoms and 2 oxygen atoms on both sides.

How to Balance Equations by Inspection

- 1) From a word equation or written statement, write the skeleton equation by replacing each name with a correct formula

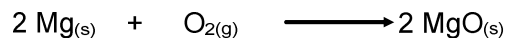
Ex. magnesium reacts with oxygen gas to produce magnesium oxide



- 2) Count the number of atoms of each type in the reactants and products.

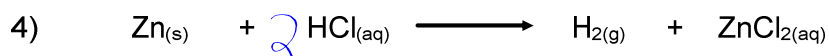
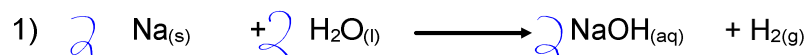
Type of atom	Reactants	Products
Mg	1	1
O	2	1

- 3) Multiply each of the formulas by the appropriate coefficients to balance the number of atoms.



HINT - Leave O and H until last

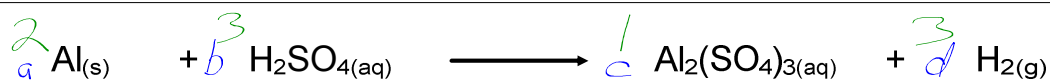
Examples



Sometimes, solving an equation by inspection can be difficult. Another alternative is to solve **algebraically**.

How to Balance Equations Algebraically

- 1) Assign a variable to each coefficient
- 2) Compare elements on reactant and product side to create linear equations
- 3) Let the variable with the largest coefficient be 1
- 4) Solve by substitution
- 5) If necessary, multiply to make whole



$$\text{Al: } a = 2c$$

$$\text{H: } 2b = 2d$$

$$\text{S: } b = 3c$$

$$\text{O: } 4b = 12c$$

$$\text{Let } \boxed{c = 1}$$

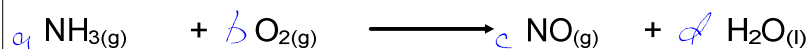
$$a = 2(1)$$

$$\boxed{a = 2}$$

$$b = 3(1)$$

$$\boxed{b = 3}$$

$$\therefore \boxed{d = 3}$$



$$\text{N: } a = c$$

$$\text{H: } 3a = 2d$$

$$\text{O: } 2b = c + d$$

$$2b = (1) + \frac{3}{2}$$

$$2b = \frac{2}{2} + \frac{3}{2}$$

$$2b = \frac{5}{2}$$

$$\boxed{b = \frac{5}{4}}$$

$$a = 1$$

$$b = \frac{5}{4}$$

$$c = 1$$

$$d = \frac{3}{2}$$

$$\text{Let } \textcircled{a = 1}$$

$$\therefore \textcircled{c = 1}$$

$$3(1) = 2d$$

$$\textcircled{d = \frac{3}{2}}$$

$$a = 4$$

$$b = 5$$

$$c = 4$$

$$d = 6$$

$$\textcircled{\times 4}$$



$$\text{C: } 3a = c$$

$$\text{H: } 6a = 2d$$

$$\text{O: } 2b = 2c + d$$

$$\text{Let } a = 1$$

$$\therefore c = 3$$

$$d = 3$$

$$2b = 2(3) + (3)$$

$$2b = 9$$

$$b = \frac{9}{2}$$

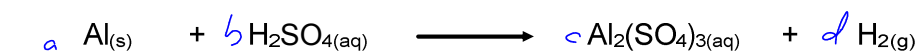
$$a = 1$$

$$b = \frac{9}{2}$$

$$c = 3$$

$$d = 3$$

$a = 2$
$b = 9$
$c = 6$
$d = 6$



$$\text{Al: } a = 2c$$

$$\text{H: } 2b = 2d$$

$$b = d$$

$$\text{S: } b = 3c$$

$$\text{O: } 4b = 12c$$

$$\text{Let } c = 1$$

$$b = 3(1)$$

$$b = 3$$

$$d = 3$$

$$a = 2(1)$$

$$a = 2$$

